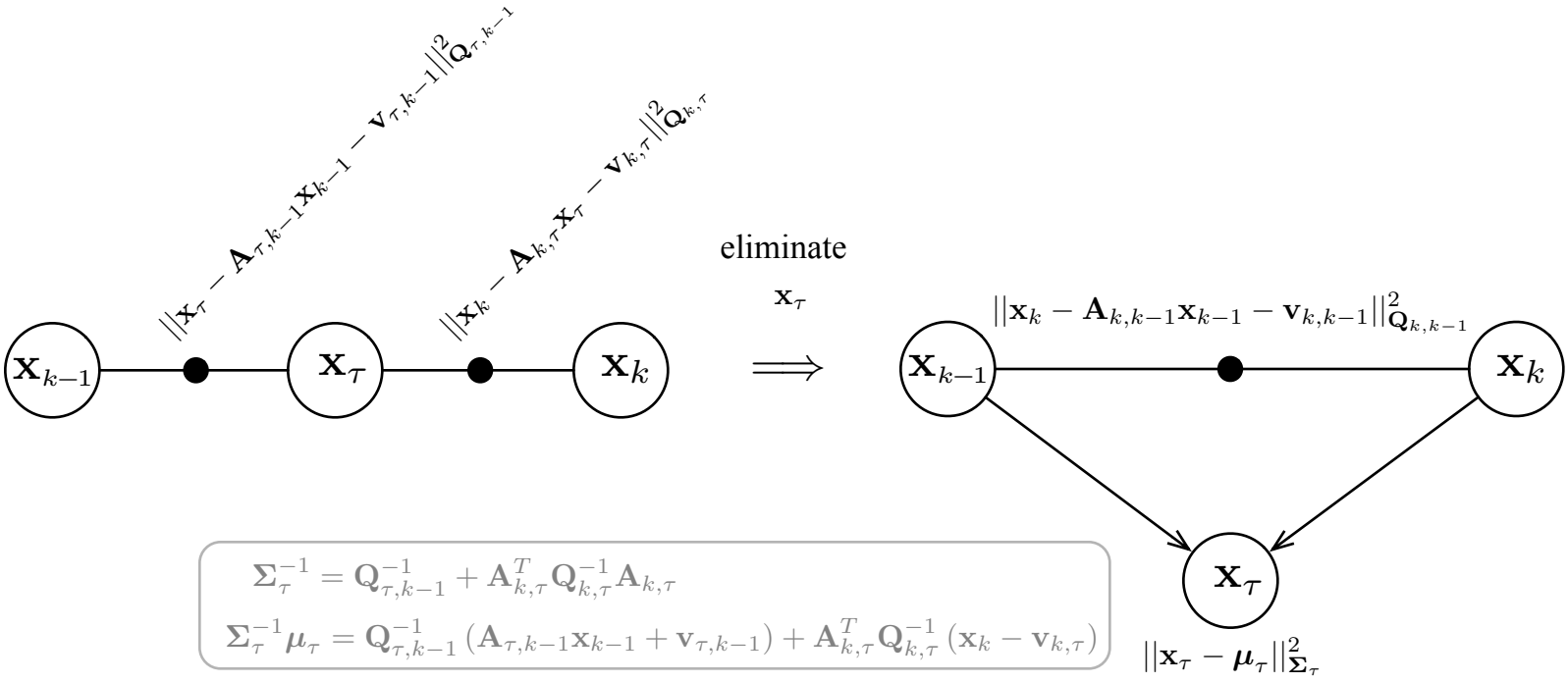

$$\boldsymbol{\Sigma}_\tau^{-1} = \mathbf{Q}_{\tau,k-1}^{-1} + \mathbf{A}_{k,\tau}^T \mathbf{Q}_{k,\tau}^{-1} \mathbf{A}_{k,\tau}$$

$$\boldsymbol{\Sigma}_\tau^{-1} \boldsymbol{\mu}_\tau = \mathbf{Q}_{\tau,k-1}^{-1} \left(\mathbf{A}_{\tau,k-1}\mathbf{x}_{k-1} + \mathbf{v}_{\tau,k-1}\right) + \mathbf{A}_{k,\tau}^T \mathbf{Q}_{k,\tau}^{-1} \left(\mathbf{x}_k - \mathbf{v}_{k,\tau}\right)$$